

Basic Transactions

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An SQL transaction is a single unit of work that groups one or more SQL operations (such as INSERT, UPDATE, or DELETE) to ensure they either all succeed or all fail together. Core Properties (ACID)

ACID properties ensure database reliability:

- Atomicity: All operations complete or the transaction rolls back.
- Consistency: Data moves between valid states, adhering to rules.
- Isolation: Concurrent transactions do not interfere with each other.
- Durability: Committed changes are permanent, even during crashes.

Transaction Control Commands

Common SQL commands:

- BEGIN TRANSACTION: Starts a new transaction.
- COMMIT: Permanently saves changes.
- ROLLBACK: Undoes changes if an error occurs.
- SAVEPOINT: Sets a point to which you can partially roll back.

Example 1: Basic Transaction with COMMIT

Insert a team and players together. If all statements succeed, commit the transaction.

```
BEGIN TRANSACTION;
```

```
INSERT INTO team VALUES (5, 'Quetta Gladiators', 'Viv Richards', 'Quetta');
```

```
INSERT INTO player VALUES ('Sarfraz Ahmed', 201, 36, 'Wicketkeeper', 75000, 5, NULL);
```

```
INSERT INTO player VALUES ('Jason Roy', 202, 33, 'Batsman', 80000, 5, 201);
```

COMMIT;

Example 2: Transaction with Conditional ROLLBACK

Update salary and ensure it does not exceed a limit.

BEGIN TRANSACTION;

UPDATE player

SET salary = salary + 50000

WHERE pid = 101;

IF (SELECT salary FROM player WHERE pid = 101) > 120000

BEGIN

ROLLBACK;

END

ELSE

BEGIN

COMMIT;

END
